UNIVERSITY OF CAPE COAST

COLLEGE OF AGRICULTURE AND NATURAL SCIENCE

SCHOOL OF PHYSICAL SCIENCES

Group 11 & 12 **PHY 101: General Physics I** **TUTORIAL (April 2021)**

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1. An object is launched at a velocity of 20m/s in a direction making an angle of 25° upward with the horizontal.
    a. What is the maximum height reached by the object?
    b. What is the total flight time of the object?
    c. Determine the range of the object (horizontal distance covered by the object)
    d. What is the magnitude of the velocity of the object just before it hits the ground?
2. A 50 g mass vibrates in S.H.M at the end of a spring. The amplitude of the motion is 12 cm, and the period is 1.70 s. Find
    a. the frequency,
    b. the spring constant,
    c. the maximum speed of the mass,
    d. the maximum acceleration of the mass.
3. Find the work done in moving an object along a straight line from (3, 2, -1) to (2, -1, 4) in a force field given by F = 4i – 3j +2k. Calculate the power dissipated if it took 4s to move the object to its final position.

ELASTICITY

4. A metal bar with square cross section (20mm x 20mm) and length L= 3.0m is subjected to an axial tensile force of 50kN. The measured elongation of the bar is ΔL= 1.5mm. Calculate the tensile stress and strain in the bar. Hence find the Young's modulus of the metal bar.

FLUIDS

5. The area of a piston of a force pump is 8cm2. What force must be applied to the piston to raise oil (ρ= 0.78g/cm3) to a height of 6.0m? Assume the upper end of the oil is open to the atmosphere. **(Ans: 37N)**
6. A hydraulic lift raises a 2 000-kg automobile when a 500-N force is applied to the smaller piston. If the smaller piston has an area of 10 cm^2, what is the cross-sectional area of the larger piston? **(Ans: 392 cm)**
7. Some species of whales can dive to depths of one kilometer. What is the total pressure they experience at this depth? (r_{sea} = 1 020 kg/m^3 and 1.01 ´ 10^5 N/m^2 = 1 ATM.). **(Ans: 100 ATM)**

KINETIC THEORY AND THERMODYNAMICS

8. A tank used for filling helium balloons has a volume of 0.300 m3 and contains 2.00 mol of helium gas at 20.0 0C. Assuming that the helium behaves like an ideal gas,

a. what is the total translational kinetic energy of the molecules of the gas? **(Ans: 7300J)**

b. What is the average kinetic energy per molecule? **(Ans: 6.07×10^{-21} J)**

9. Helium gas occupies a volume of 0.04 m3 at a pressure of 2 x 105 Pa (Nm-2) and temperature 300 K. Calculate
    a. the mass of helium, **(Ans: 12.8 g)**
    b. the r.m.s speed of its molecules, **(Ans: 1369 ms^{-1})**
10. What is the final temperature of a mixture of 100 g of water at 90 ℃ and 600g of water at 20 ℃? Specific Heat Capacity for water is 4186 J/kg℃. **(Ans: 30 ℃)**
11. 2.00 L of a gas is known to be at a temperature of 27 ℃ and at 740 mmHg pressure. What is its volume at standard atmospheric pressure of 760 mmHg? **(Ans: 1.95 L)**